E-Commerce SQL Data Analysis

Industrial Training Project – DecodeLabs

SQL Project Report

Submitted By

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Domain

Data Analytics

Technology

MySQL • SQL • Microsoft Excel • Git • GitHub

Organization

DecodeLabs Industrial Training

Year

2026

Table of Contents

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1. Introduction

In today's competitive business environment, organizations generate vast amounts of transactional data every day. SQL (Structured Query Language) is one of the most important tools used by Data Analysts to retrieve, filter, analyze, and summarize data stored in relational databases.

This project focuses on analyzing an E-Commerce sales dataset using MySQL. Various SQL queries were executed to understand customer behavior, product performance, revenue generation, payment preferences, order status, and marketing effectiveness. The project demonstrates how raw business data can be transformed into meaningful insights that support informed decision-making.

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2. Project Objective

The primary objectives of this project are:

- Create and manage a relational database.
- Import an E-Commerce dataset into MySQL.
- Perform SQL-based data analysis.
- Apply filtering, grouping, sorting, and aggregation.
- Generate business insights from sales data.
- Develop practical SQL skills for real-world business scenarios.

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3. Dataset Description

The dataset contains approximately **1,200** e-commerce transaction records.

Attributes

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice

Each row represents one customer transaction containing order details, payment information, product details, and total sales amount.

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4. Database Design

Database Name:

EcommerceDB

Table Name:

EcommerceSales

The database consists of a single transactional table designed to store customer order information efficiently. Appropriate data types such as VARCHAR, DATE, INT, and DECIMAL were used to maintain data integrity.

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5. Tools & Technologies Used

- MySQL Server
- MySQL Workbench
- SQL
- Microsoft Excel
- Git
- GitHub
- Visual Studio Code

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6. SQL Concepts Used

The project demonstrates the practical use of several SQL concepts:

- CREATE DATABASE
- CREATE TABLE
- INSERT INTO
- SELECT
- DISTINCT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()
- ROUND()
- CASE
- Subqueries

- LIMIT

These SQL operations were used to answer business questions and extract meaningful information from the dataset.

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7. Data Import Process

The dataset was imported into MySQL Workbench using the Table Data Import Wizard. After importing the data, validation checks were performed to verify:

- Total records
- Missing values
- Duplicate Order IDs
- Date range consistency
- Data type correctness

This ensured that the dataset was clean and ready for analysis.

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8. Business Analysis

Multiple SQL queries were executed to answer important business questions.

The analysis covered:

- Overall Revenue
- Total Orders
- Average Order Value
- Product Performance
- Customer Spending
- Customer Purchase Frequency
- Payment Method Analysis
- Coupon Performance
- Referral Source Analysis
- Monthly Revenue
- Order Status Distribution
- Shopping Cart Analysis
- Revenue Contribution

These analyses provide valuable information for improving business performance.

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9. Key SQL Queries

The following categories of SQL queries were implemented:

Basic Queries

- Display all records
- Filter completed orders
- Sort products by revenue
- Retrieve cancelled orders

Intermediate Queries

- Revenue by Product
- Revenue by Payment Method
- Orders by Referral Source
- Monthly Revenue
- Product Sales Analysis

Advanced Queries

- Highest Spending Customers
- Revenue Contribution Percentage
- Cancellation Rate
- Return Rate
- Customer Purchase Frequency
- Average Customer Spending
- Coupon Performance Analysis

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10. Business Insights

The SQL analysis generated several valuable business insights:

- High-performing products contribute the largest share of total revenue.
- A small percentage of customers generate a significant portion of overall sales.
- Average Order Value indicates customer purchasing behavior.
- Some products require better marketing due to lower sales.
- Preferred payment methods should be optimized for faster checkout.
- Coupon performance identifies successful promotional campaigns.
- Referral source analysis highlights the most effective marketing channels.
- Monthly sales trends support inventory planning.
- Cancellation and return analysis identify opportunities for operational improvement.
- Shopping cart analysis helps improve cross-selling and upselling strategies.

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11. Conclusion

This project successfully demonstrates the application of SQL for business analytics using a real-world E-Commerce dataset.

By applying SQL queries such as filtering, grouping, aggregation, sorting, subqueries, and conditional logic, meaningful business intelligence was extracted from raw transactional data.

The project strengthened practical knowledge of SQL while showcasing the ability to solve real business problems using data analysis techniques. The insights generated through this project can help organizations improve operational efficiency, customer satisfaction, marketing effectiveness, and revenue growth.

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12. Future Scope

The project can be enhanced by incorporating additional technologies and advanced analytics:

- Power BI Dashboard
- Tableau Dashboard
- Python Data Analysis
- Predictive Sales Forecasting
- Customer Segmentation
- Inventory Optimization
- Machine Learning Models
- Interactive Business Intelligence Reports

These improvements would provide deeper insights and support data-driven strategic decision-making.

References

- MySQL Documentation
- SQL Language Reference
- Microsoft Excel Documentation
- DecodeLabs Industrial Training Material
- GitHub Documentation

Acknowledgement

I sincerely thank **DecodeLabs** for providing the industrial training opportunity and practical project guidance. This project enhanced my understanding of SQL, database management, and business analytics while allowing me to apply theoretical concepts to real-world scenarios.

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2026